

Exercise 4 – Case Study: zoom

Learning objectives:

1. Understand parameterized testing
2. Understand the basics of test planning and risk assessment

Task 1 – Parameterize the Triangle Software Tests

Our triangle software from Exercise 2 contains a large number of test cases. In this exercise, we make the test more readable and maintainable.



- 1.1.) Read up on parameterized tests for the JUnit framework:

<https://junit.org/junit5/docs/current/user-guide/#writing-tests-parameterized-tests>

- 1.2.) Rework your tests into as few tests as possible (ideally just 1), while still covering all test cases. Pass the test inputs (test data) and expected outputs into the test method.

Annotate the Method using the `@ParameterizedTest` annotation.

Task 2 – Test Planning and Test Strategy

In your role as test manager, you are responsible for the test of the application “zoom”.

- 2.1.) Define test objectives, scope and risks of testing.

- 2.2.) Outline: Which test objects you will consider in your test process?

- 2.3.) Determine what your test team should look like. Justify your resource requirements!

- 2.4.) Given three defined test intensity levels, depending on criticality. Assign the features to a test intensity level depending on risk. How would you proceed?

Write down your initial thoughts and then look at the next page.

2.4.) Given three defined test intensity levels, depending on criticality. Assign the features to a test intensity level depending on risk. How would you proceed?

		Probability of Failure Occurrence		
		high	medium	low
Damage	high	I1	I1	I1
	medium	I1	I2	I2
	low	I2	I3	I3

- I1 High test intensity
- I2 Medium test intensity
- I3 Low test intensity

What features would you write into which cell?

		Probability of Failure Occurrence		
		high	medium	low
Damage	high			
	medium			
	low			

- I1 High test intensity
- I2 Medium test intensity
- I3 Low test intensity

2.5.) Prioritize loosely based on test intensity. With which “cells” would you start?

Enumerate your cells according to priority (1 through 9).

		Probability of Failure Occurrence		
		high	medium	low
Damage	high	1		
	medium			
	low			9

Task 3 – Test Analysis

In your role as test manager, you are responsible for the test of the application “zoom”.

3.1.) Part of the test team is currently reviewing the test basis. Your project manager is very skeptical and cannot fully understand the benefits (compared to the effort involved). What are the arguments in favor of your chosen test process activities?

3.2.) Outline the feature “Meeting participant” using at least 5 potential test conditions.

3.3.) Select one or more test conditions and intuitively derive 5 test cases. Define preconditions, postconditions and possible test steps. What test data is required?